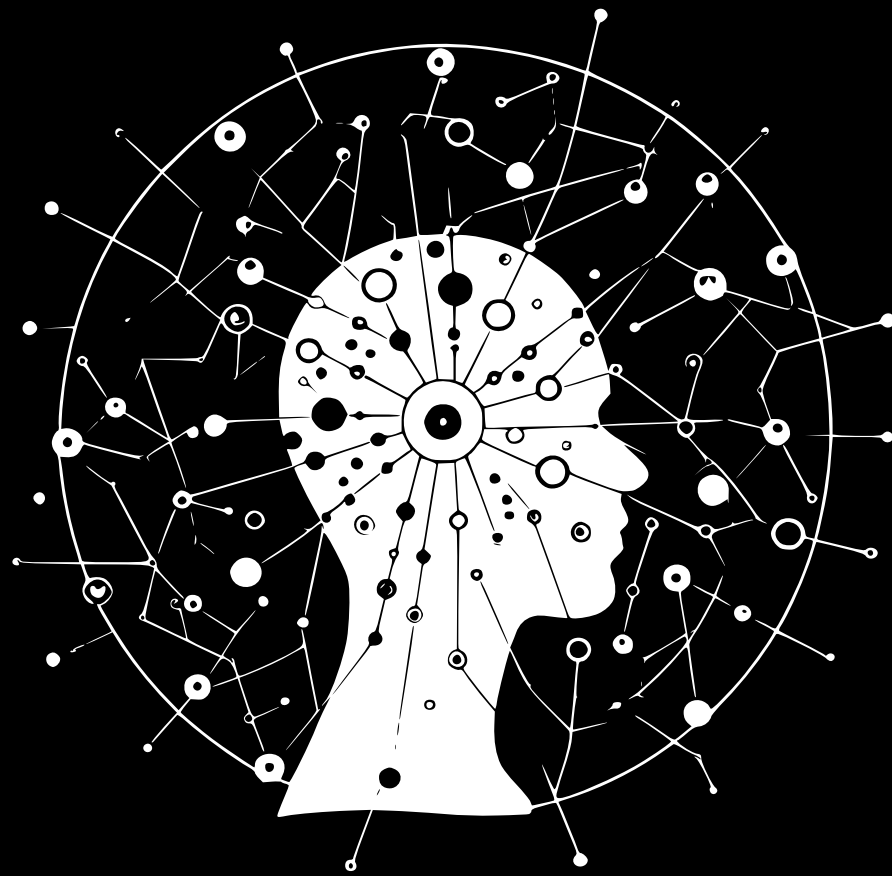


Summer Lectures 2026



Lecture 1



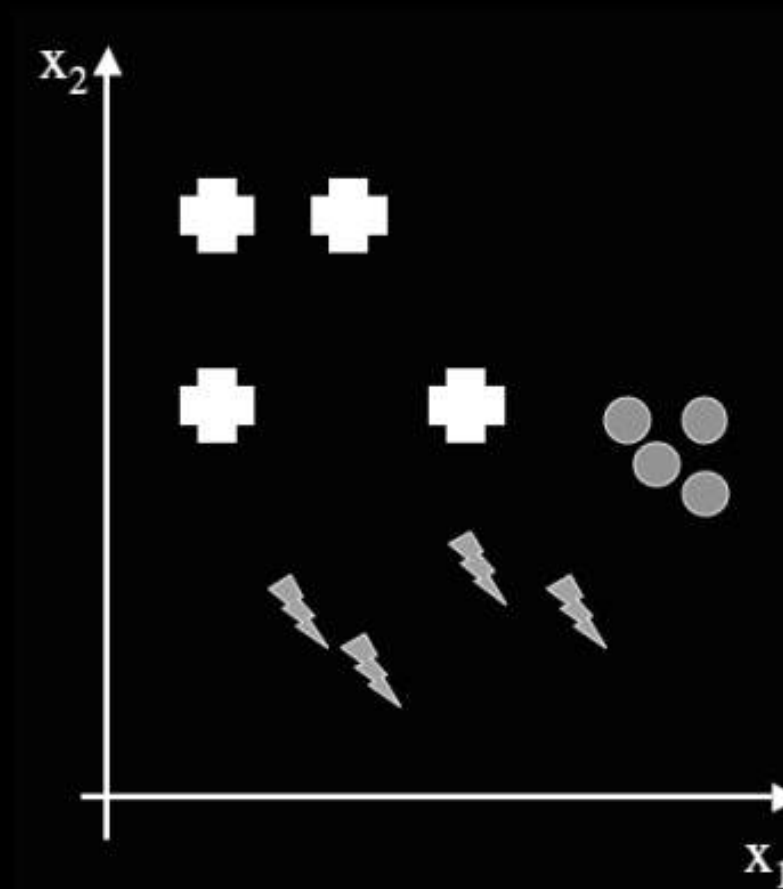
“Around the world, computers
captures and store terabytes of
data everyday”



“The challenge is to do
something useful with this
data”

Size and complexity of data

x_1	x_2	Class
0.1	1	1
0.15	0.2	2
0.48	0.6	3
0.1	0.6	1
0.2	0.15	2
0.5	0.55	3
0.2	1	1
0.3	0.25	2
0.52	0.6	3
0.3	0.6	1
0.4	0.2	2
0.52	0.5	3



Size and complexity. Adapted from Machine Learning: An Algorithmic Perspective (2nd ed., p. 3), by S. Marsland, 2015, CRC Press. Copyright 2015 by Taylor & Francis Group, LLC.

Higher Dimensions



Higher Dimensions. Adapted from Machine Learning: An Algorithmic Perspective (2nd ed., p. 3), by S. Marsland, 2015, CRC Press. Copyright 2015 by Taylor & Francis Group, LLC.

What is learning?

Key Concept: Learning from data

ANALIZING ANIMAL LEARNING



IMPORTANT ASPECTS:

- REMEMBERING
- ADAPTING
- GENERALIZING

RECOGNIZING SIMILARITY BETWEEN
DIFFERENT SITUATIONS



Adjusting and adapting to new
circumstances through
experience

“Learning give us flexibility in our life”



How is this process in general?

- 1.- We see data (an Input)
- 2.- Trying out a particular action (an output)
- 3.- Did it work? If not, try something new.

Other bits to intelligence

- Reasoning
- Logical deduction

What is AI?

The overarching field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence.



What is ML?

A specific subset of AI that focuses on the development of algorithms that allow computers to "learn" from data. Instead of being explicitly programmed for every specific task, an ML model identifies patterns within a dataset to make predictions or decisions.



What is DL?

A specialized evolution of Machine Learning inspired by the structure and function of the human brain's neural networks.



History of DL

- 1940 - 1960: Cybernetics
- 1980 - 1990: Connectionism
- 2006: Deep Learning



Remember...

Learning: Getting better at some task through practice

So, How does the computer know is getting better?

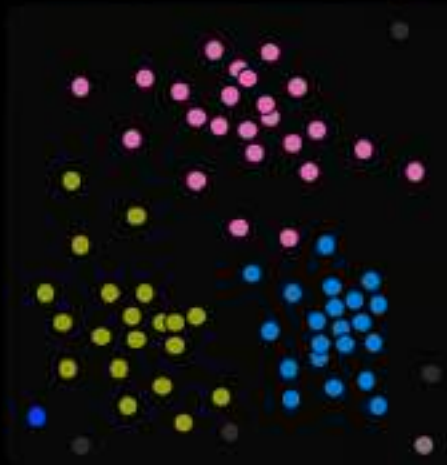
- Give a few correct answers
- Tell if answers are correct or not (or a score of how correct the answer is)
- We might not have any correct answers: Find inputs that have something in common



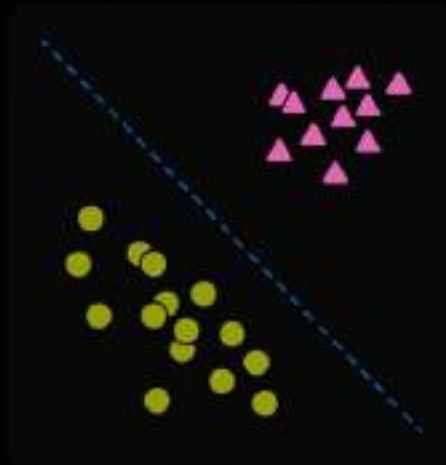
Types of ML

Machine Learning

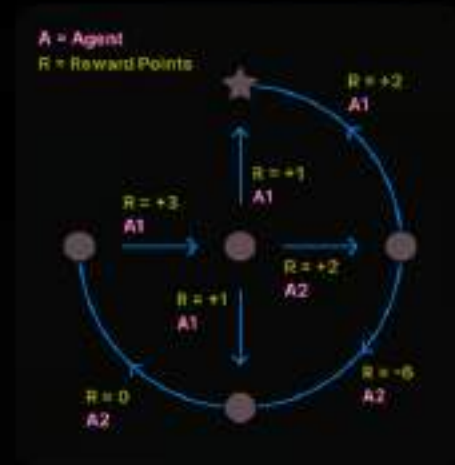
Unsupervised Learning



Supervised Learning



Reinforcement Learning



- Supervised learning

Targets/labels: Learning from exemplars

- Unsupervised Learning

Correct responses are not provided: Density estimation

- Reinforcement learning

Learning with a critic

- Evolutionary learning

Inspired in how organism adapt to improve their survival rates

Adapted from <https://fpt.ai/wp-content/uploads/2025/01/generative-ai-vs-machine-learning-2.png>

Supervised Learning

Dataset → training data

Input and target data



x_i



t_i

Supervised Learning in ML

Input



Annotations

These are
apples



Model



It's Apples

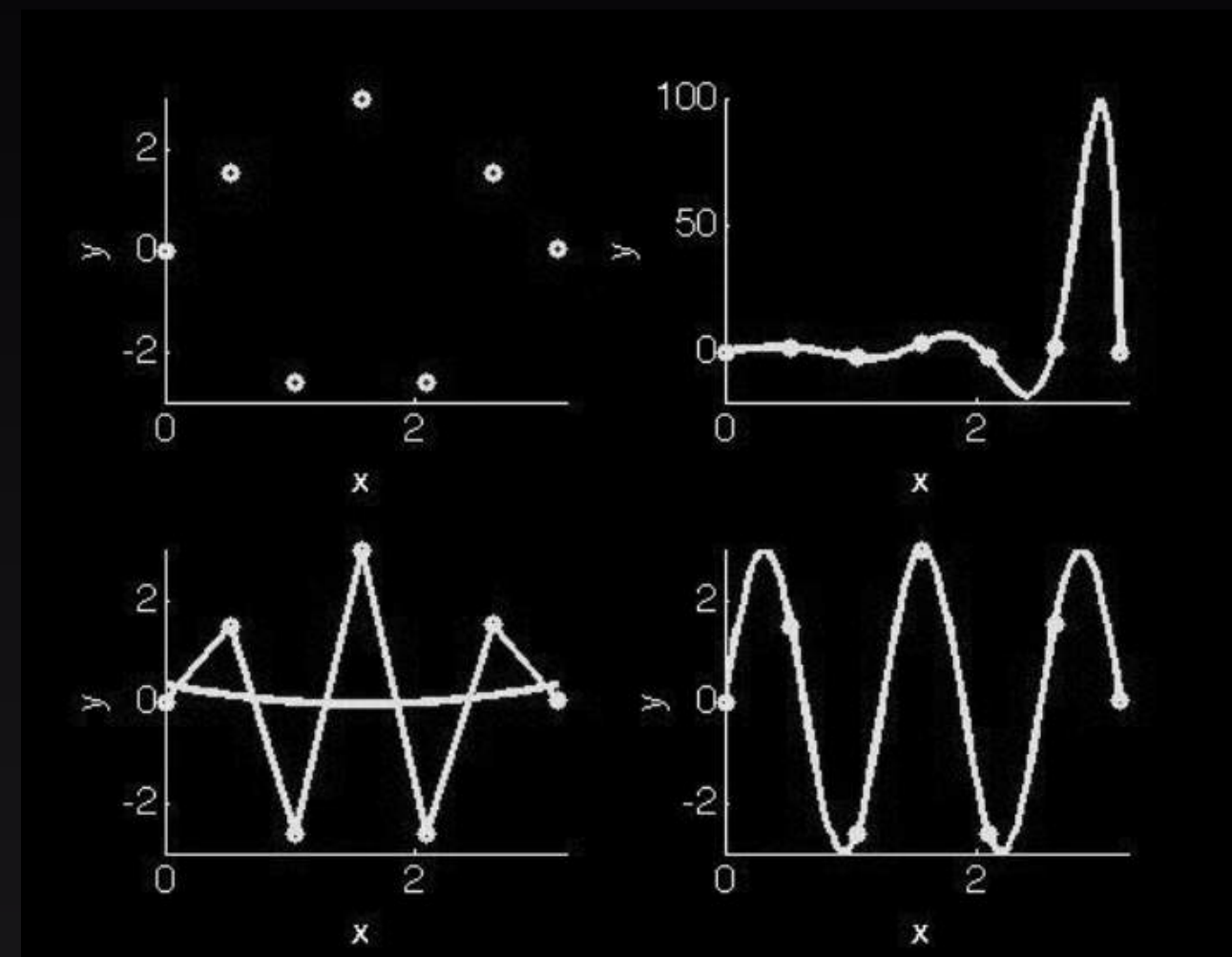
Prediction

Adapted from https://miro.medium.com/v2/resize:fit:1400/format:webp/0*ycV8b9yM70FLQpf1.jpg

Regression

Fit a mathematical function describing a curve so that the curve passes as close as possible to all of the datapoints.

It is generally a problem of function approximation or interpolation



Regression. Adapted from Machine Learning: An Algorithmic Perspective (2nd ed., p. 7), by S. Marsland, 2015, CRC Press. Copyright 2015 by Taylor & Francis Group, LLC.

The problem is to work out what function to choose!

Classification

Taking input vectors and deciding which of N classes they belong to, based on training from examples of each class

There are two constraints that are not necessarily realistic:

- *Fuzzy Classifiers*
- *Novelty detection*



Important: it is discrete!

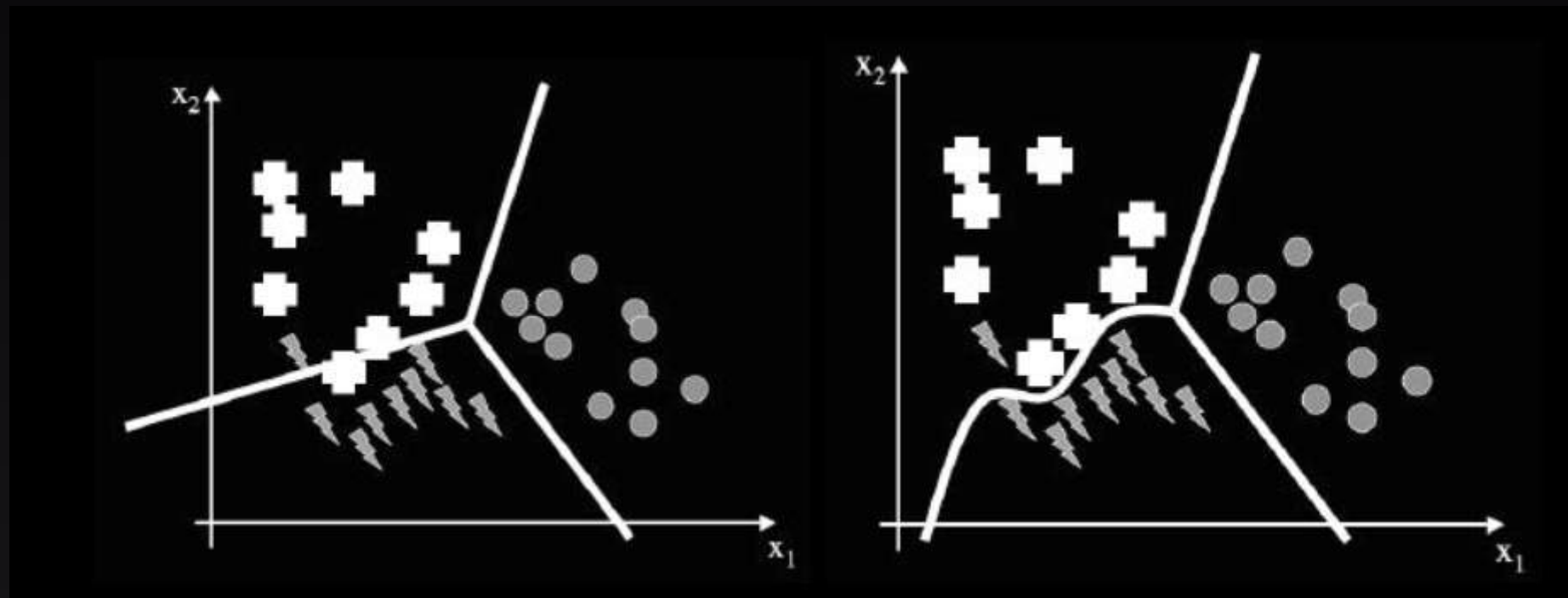


“The question of which
features to choose is not
always an easy one”

Curse of dimensionality

“If the number of input dimensions grows the number of datapoints required increases faster”

Decision boundaries



In essence they aimed
to do the same thing:
Find decision
boundaries

Decision Boundaries. Adapted from Machine Learning: An Algorithmic Perspective (2nd ed., p. 9), by S. Marsland, 2015, CRC Press. Copyright 2015 by Taylor & Francis Group, LLC.

The Machine Learning Process

- Data collection and preparation
 - Feature selection
 - Algorithm choice
 - Parameters and model selection
 - Training
 - Evaluation



Neurons, Neural Networks and Linear regression



“If we can understand how the brain works, then there might be things in there for us to copy and use for our machine learning systems”.

The brain: A big enigma

How does human brain work?

Why is it so impressive?

Neurons: the basic units



Potential membrane

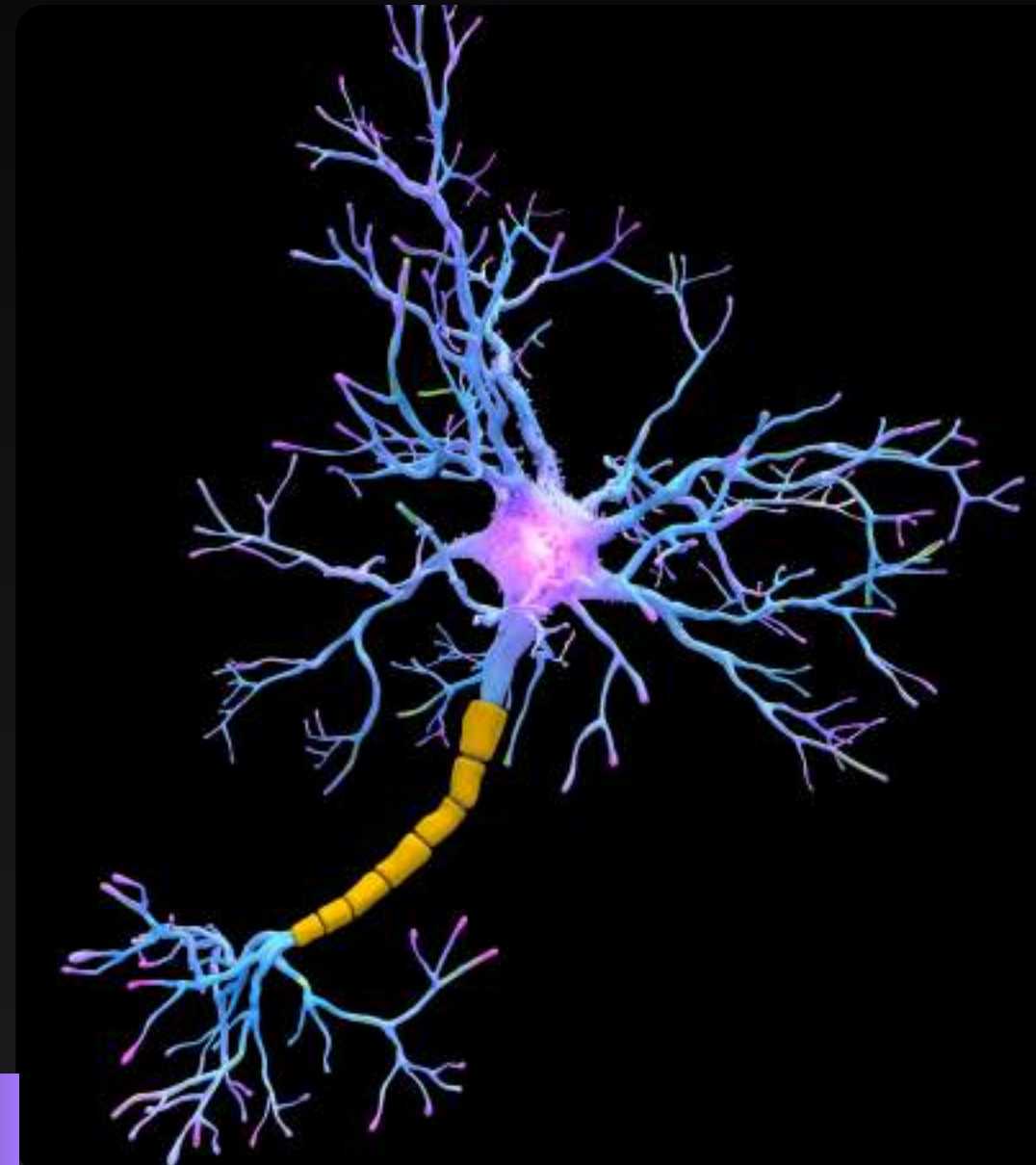
Keyword: Threshold

Neuron spikes or fires pulses of fixed strength and duration

→ axon

After being fired there's a Refractory period

Even though it is a very simple computation, all together acts as a massive parallel computer



Strong AI

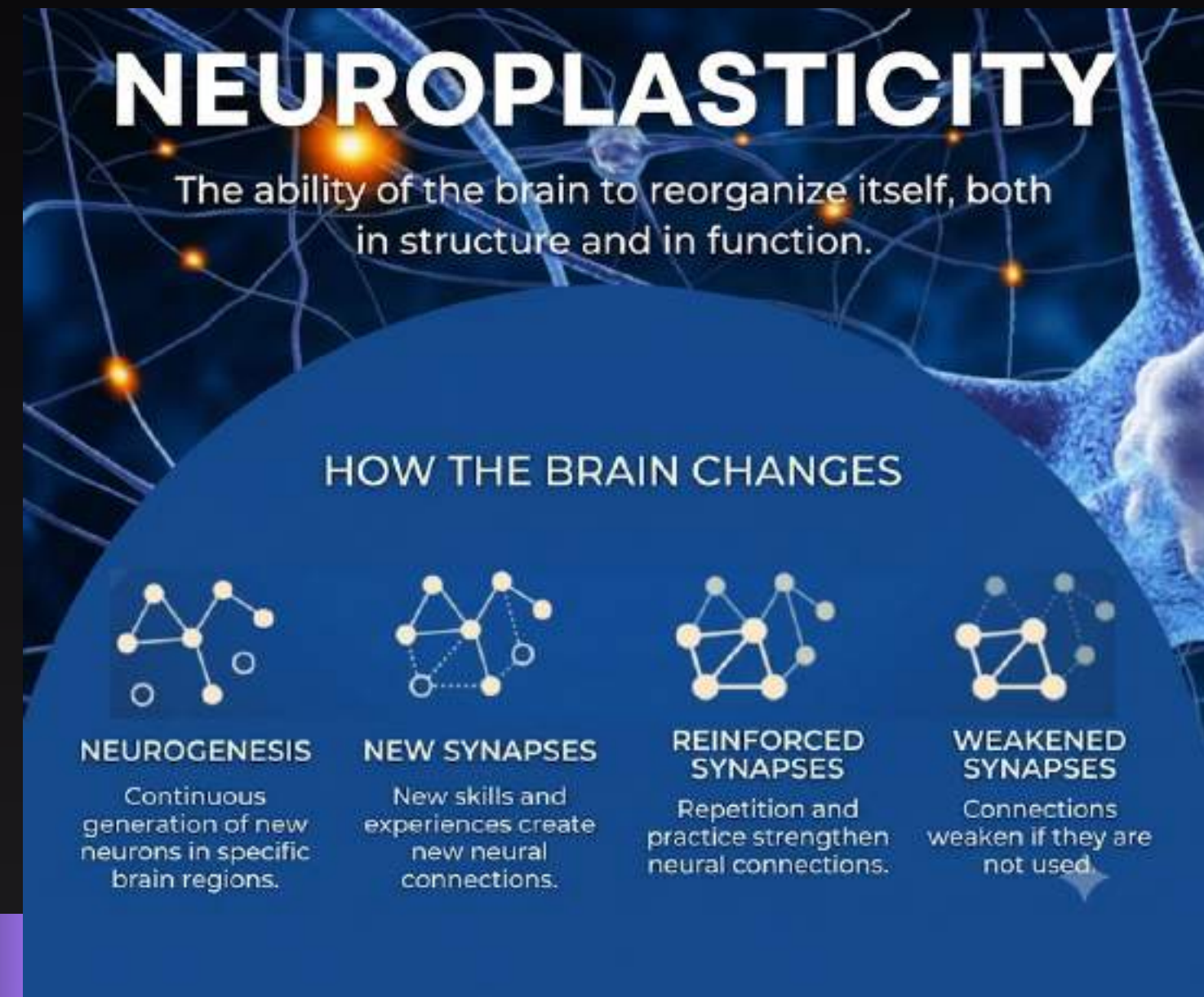
“We should be able to model it
inside a computer and end up
with animal or human
intelligence inside a computer”

How does learning occur?

Keyword: Threshold

NEUROPLASTICITY

- Strength of synaptic connections modifies
- Or new connections are created



Hebb's Rule

If two neurons consistently fire simultaneously, the any connection between them will change in strenght, becoming stronger

- Classical conditioning (Pavlov)



McCulloch and Pitt's Neurons

How do we study neurons?

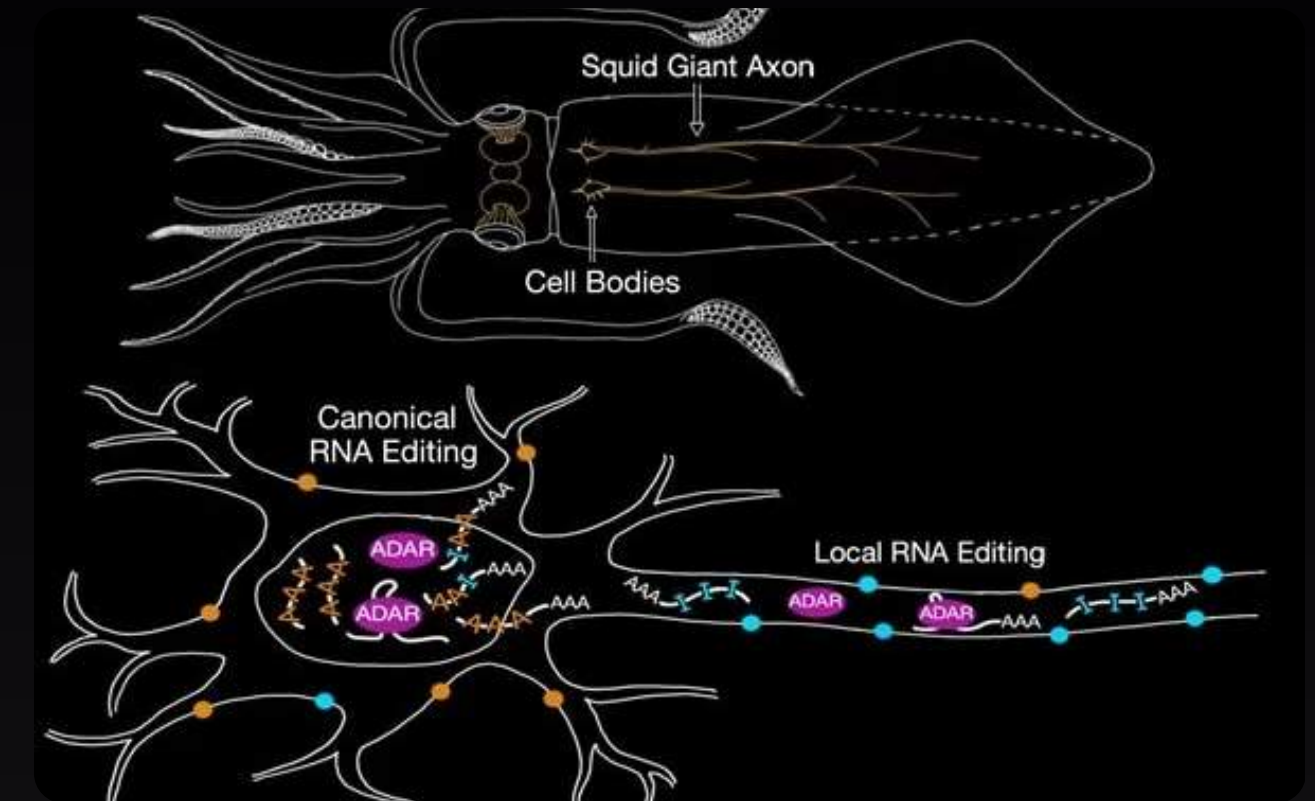
The problem

- Human neurons are too small

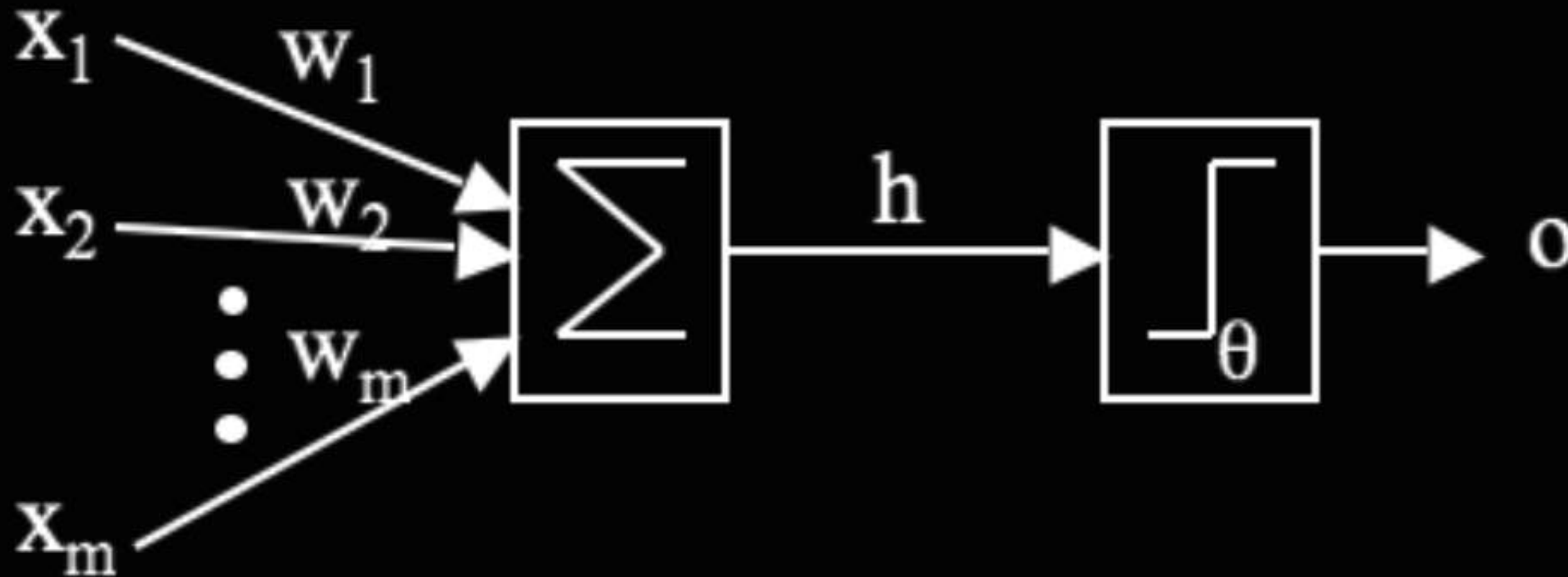


Hodgkin and Huxley (Nobel Prize)

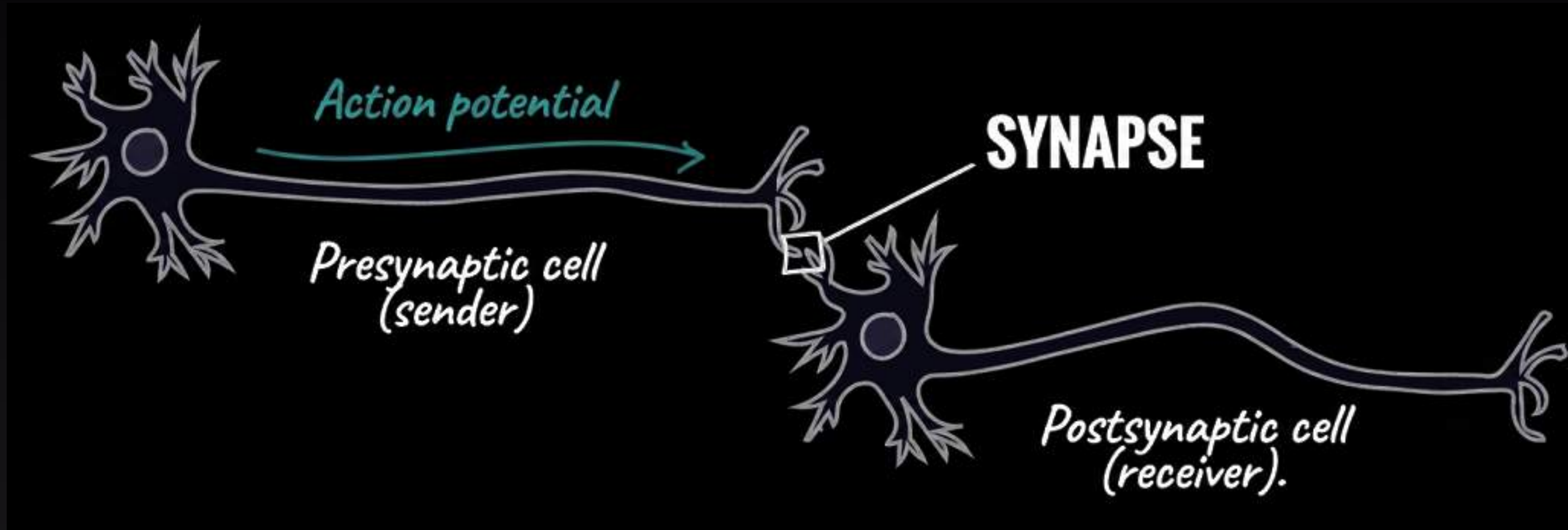
Neurons from the giant squid (1952)



The perceptron



The perceptron. Adapted from Machine Learning: An Algorithmic Perspective (2nd ed., p. 41), by S. Marsland, 2015, CRC Press. Copyright 2015 by Taylor & Francis Group, LLC.



The analogy

w_i Weighted inputs → Synapses

Σ Adder → Membrane of the cell that collects electrical charge

$\sigma(h)$ Activation function → Decides whether the neuron fires for the current input



Limitations of the model

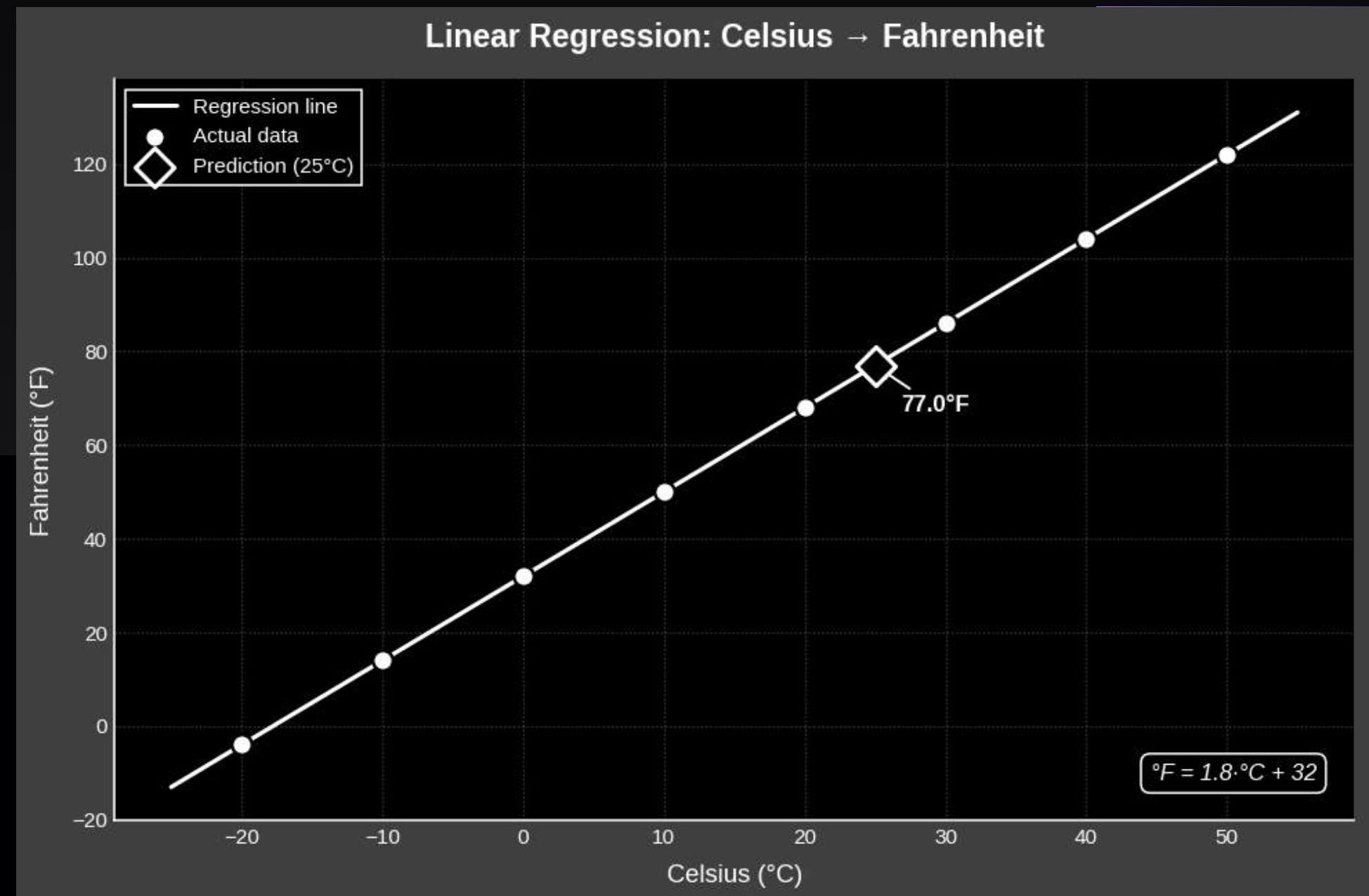
How realistic is it?

- Inputs are summed linearly
- w_i can be negative or positive
- Single output

Simple Linear Regression

$$\hat{y} = mx + b$$

Linear regression is a statistical technique used to model the relationship between two or more variables by fitting a straight line to the data. Its main purpose is to predict a continuous value based on another.





Linear Algebra, when numbers are not enough

Abelian Group

An Abelian group (or commutative group) is a set equipped with a binary operation that satisfies the commutative property, in addition to the standard group axioms: closure, associativity, the existence of an identity element, and the existence of a symmetric element or inverse.

$$\forall a, b \in G \implies a \star b \in G$$

$$\forall a, b, c \in G \implies (a \star b) \star c = a \star (b \star c)$$

$$\exists e \in G \mid \forall a \in G, a \star e = e \star a = a$$

$$\forall a \in G, \exists a^{-1} \in G \mid a \star a^{-1} = a^{-1} \star a = e$$

$$\forall a, b \in G, a \star b = b \star a$$



Niels Henrik Abel

Field

$$(F, +, \cdot)$$

A field is a triple $(F, +, \cdot)$, where F is a nonempty set, and $+$ (addition) and $\cdot$ (multiplication) are two binary operations on F that satisfy the following three fundamental conditions:

$(F, +)$ is an abelian group

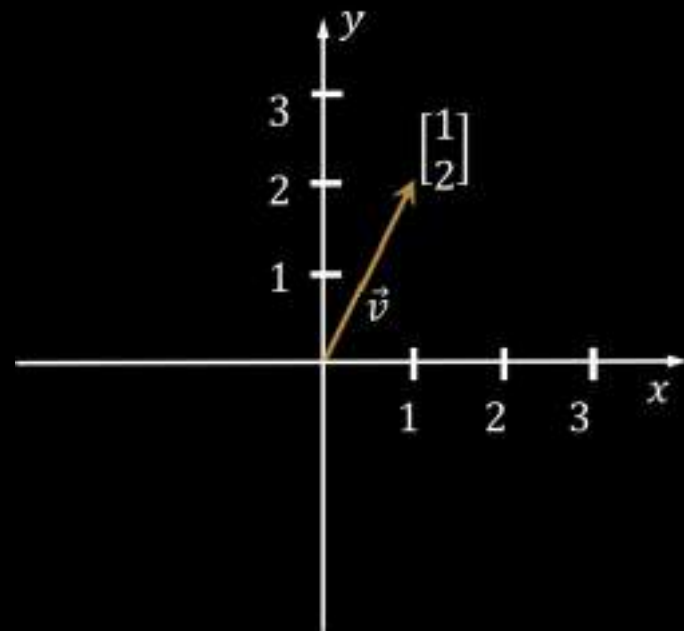
$(F \setminus \{0\}, \cdot)$ is an abelian group

Distributive Property



Vector Space

A **vector space** V over a field $\mathbb{F}$ is a triple $(V, +, \cdot)$ where V is a non-empty set and the operations of vector addition $(+ : V \times V \rightarrow V)$ and scalar multiplication $(\cdot : \mathbb{F} \times V \rightarrow V)$ satisfy the following axioms for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{F}$:



- **Additive properties:** $(V, +)$ is an abelian group.
- **Distributivity:** $\alpha(u + v) = \alpha u + \alpha v$ and $(\alpha + \beta)u = \alpha u + \beta u$.
- **Compatibility:** $\alpha(\beta u) = (\alpha\beta)u$.
- **Identity:** $1_{\mathbb{F}}u = u$.



Vectors



A **vector** v is an element of a set V known as a vector space. Formally, it is an object that satisfies the vector space axioms, allowing for vector addition and scalar multiplication. In the context of a coordinate space $\mathbb{F}^n$, a vector is an ordered n -tuple: $v = (x_1, x_2, \dots, x_n)$.

$$P(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{bmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{pmatrix}$$

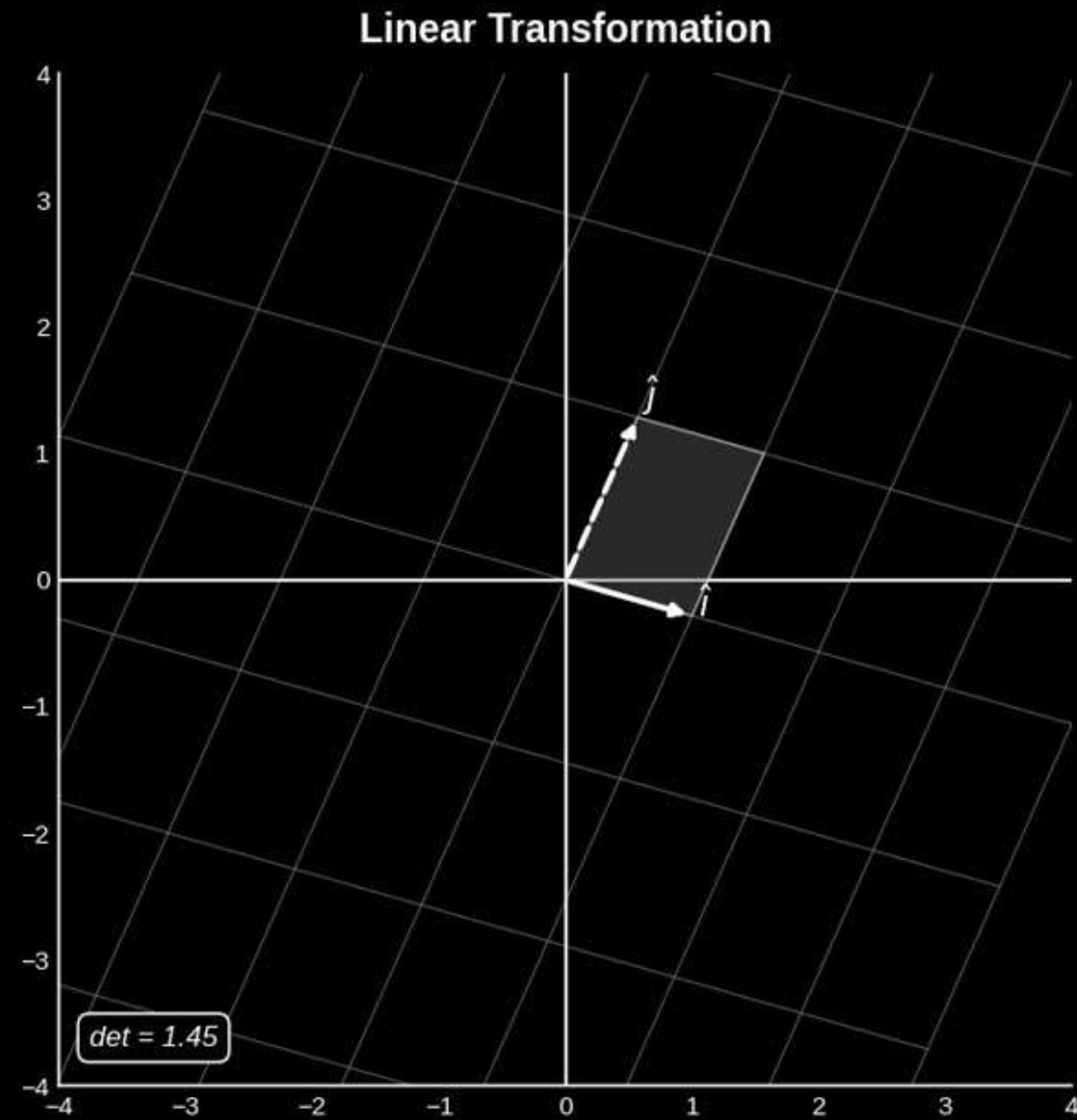
Linear Transformation

A **linear transformation** (or linear map) $T : V \rightarrow W$ is a morphism in the category $\mathbf{Vect}_{\mathbb{F}}$ that preserves the algebraic structure of the spaces:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v) \quad \forall u, v \in V, \forall \alpha, \beta \in \mathbb{F}$$

- **Kernel:** $\ker(T) = \{v \in V \mid T(v) = 0_W\}$.
- **Image:** $\text{Im}(T) = \{T(v) \mid v \in V\}$.
- **Rank-Nullity Theorem:** $\dim(V) = \dim(\ker(T)) + \dim(\text{Im}(T))$.





Basis

Let V be a vector space over a field $\mathbb{F}$. A subset $\mathcal{B} = \{v_1, v_2, \dots, v_n\} \subseteq V$ is a **basis** of V if it satisfies two conditions:

1. **Linear Independence:** The set $\mathcal{B}$ is linearly independent. That is, the only solution to the equation:

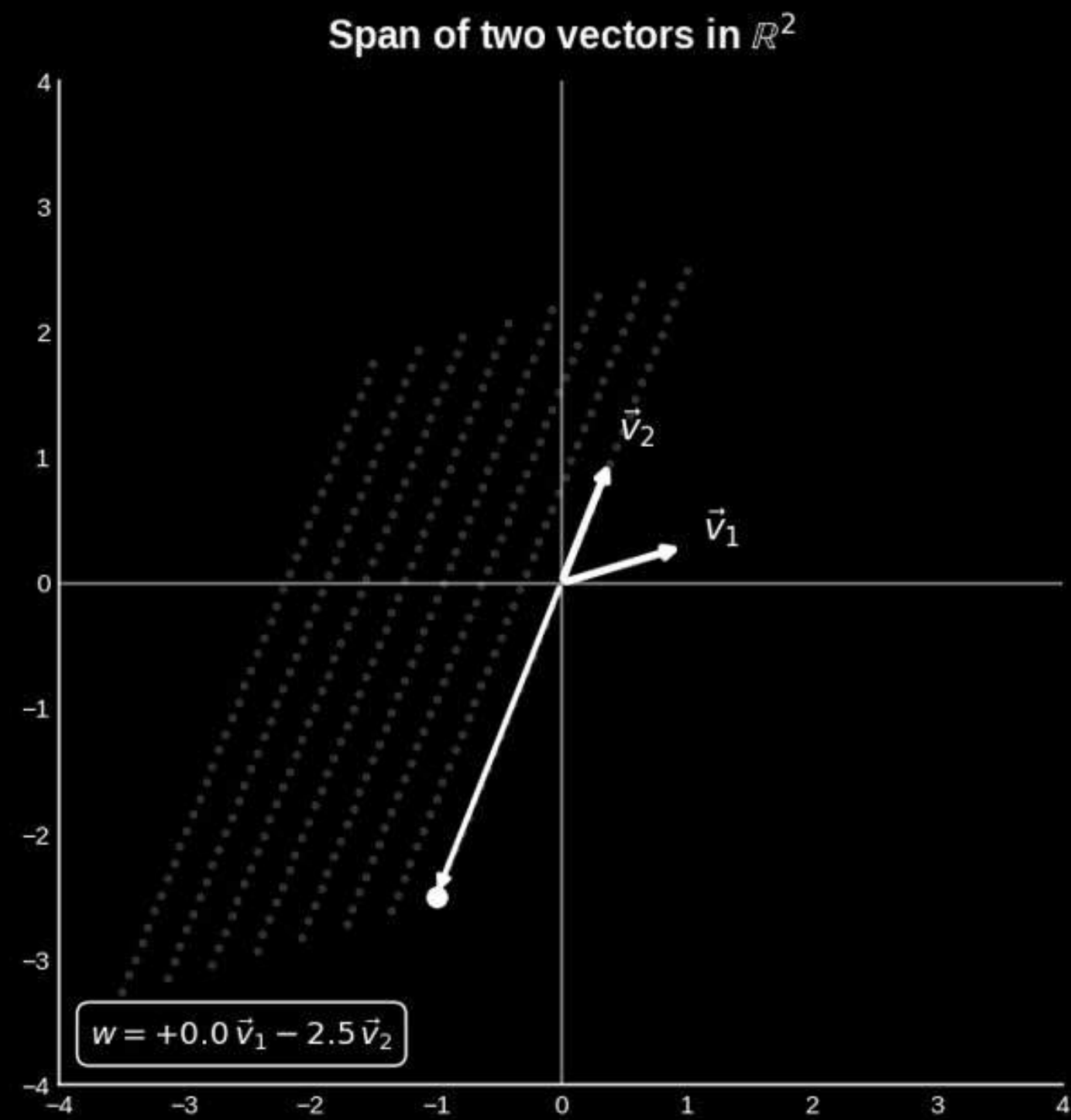
$$\sum_{i=1}^n \alpha_i v_i = 0_V$$



2. **Spanning Property (Generating Set):** The set $\mathcal{B}$ spans V . That is, every vector $v \in V$ can be expressed as a linear combination of the elements of $\mathcal{B}$:

$$V = \text{span}(\mathcal{B}) = \left\{ \sum_{i=1}^n c_i v_i \mid c_i \in \mathbb{F} \right\}$$





Matrices

Formally, a **matrix** $A \in \mathbb{F}^{m \times n}$ is the representation of a linear map $T : V \rightarrow W$ (where $\dim(V) = n$ and $\dim(W) = m$) with respect to chosen ordered bases $\mathcal{B}$ and $\mathcal{C}$. It is an $m \times n$ rectangular array of scalars $a_{i,j}$ such that the j -th column is the coordinate vector of $T(v_j)$ with respect to the basis $\mathcal{C}$.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$



Linear Functional

A **linear functional** is a special case of a linear transformation where the codomain is the field of scalars: $f : V \rightarrow \mathbb{F}$. The set of all such functionals is the **dual space** $V^* = \text{Hom}(V, \mathbb{F})$.

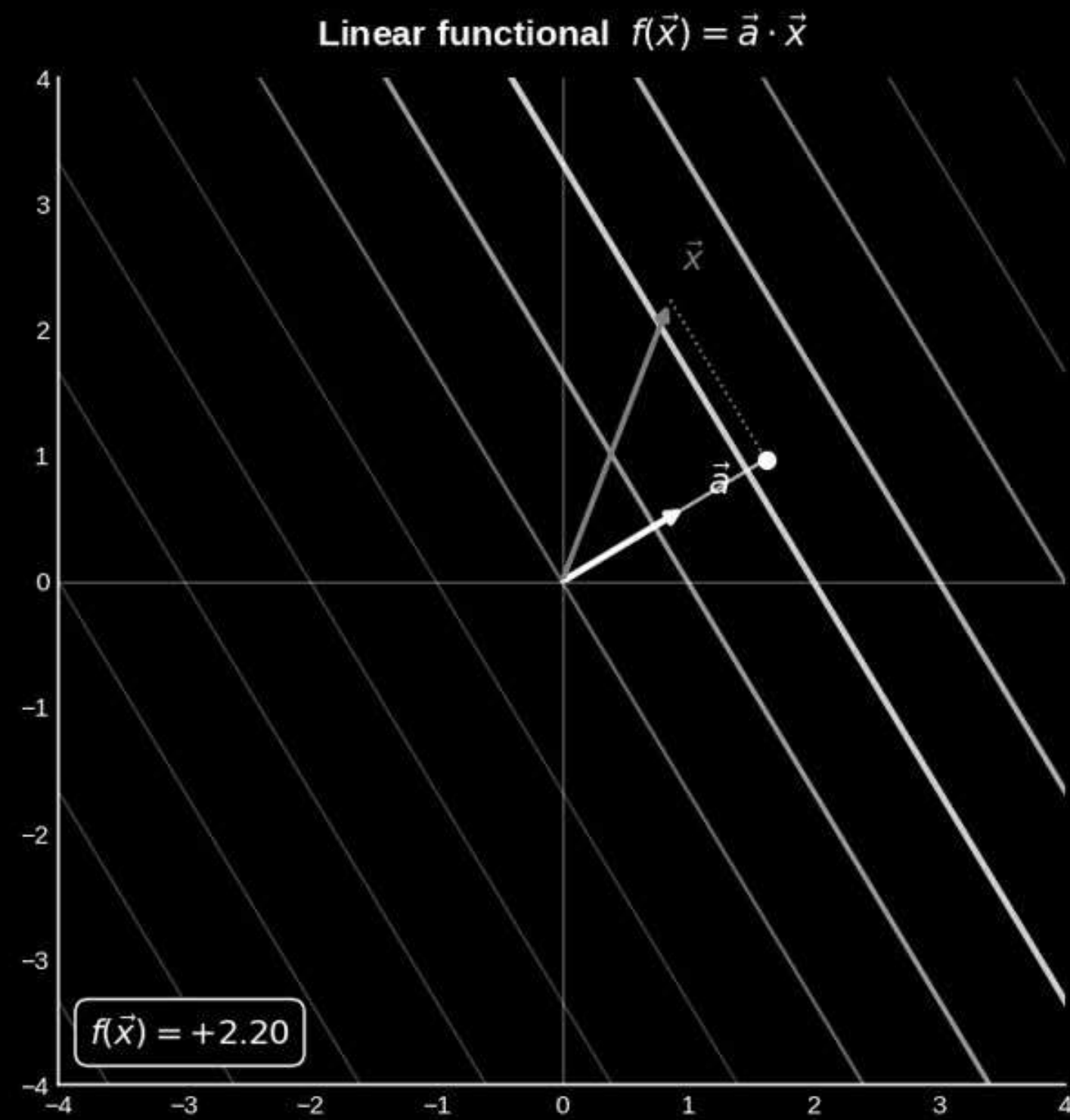


Inner Product

An inner product on a real vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying:

- **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$.
- **Linearity (in the first argument):** $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$.
- **Positive-definiteness:** $\langle u, u \rangle \geq 0$, and $\langle u, u \rangle = 0 \iff u = 0$.





Fundamental Theorem: Basis and Coordinates

Theorem (Existence and Uniqueness of Coordinates): Let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . For every $u \in V$, there exists a unique set of scalars $\{c_1, c_2, \dots, c_n\} \subset \mathbb{F}$ such that:

$$u = \sum_{i=1}^n c_i v_i$$

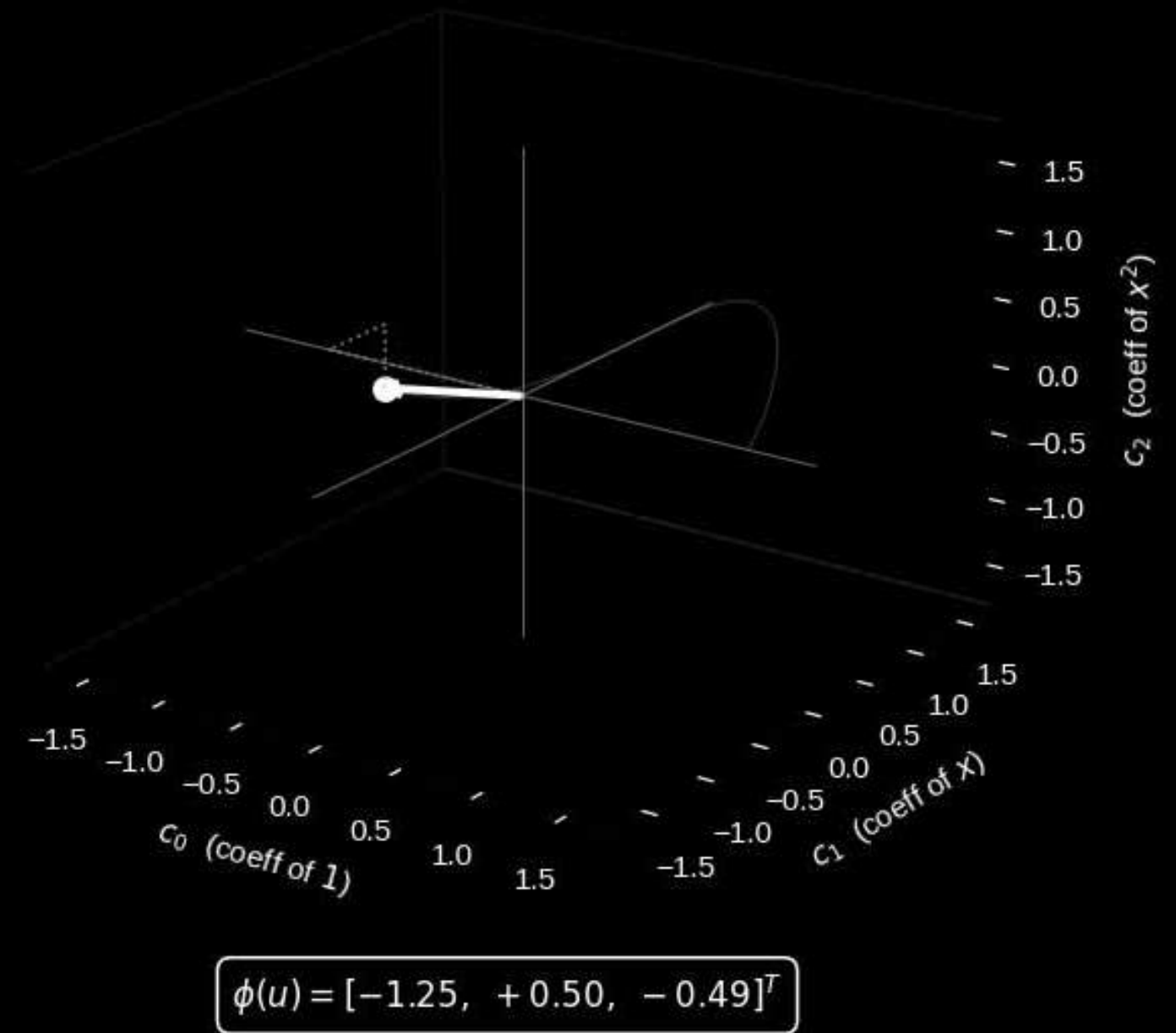
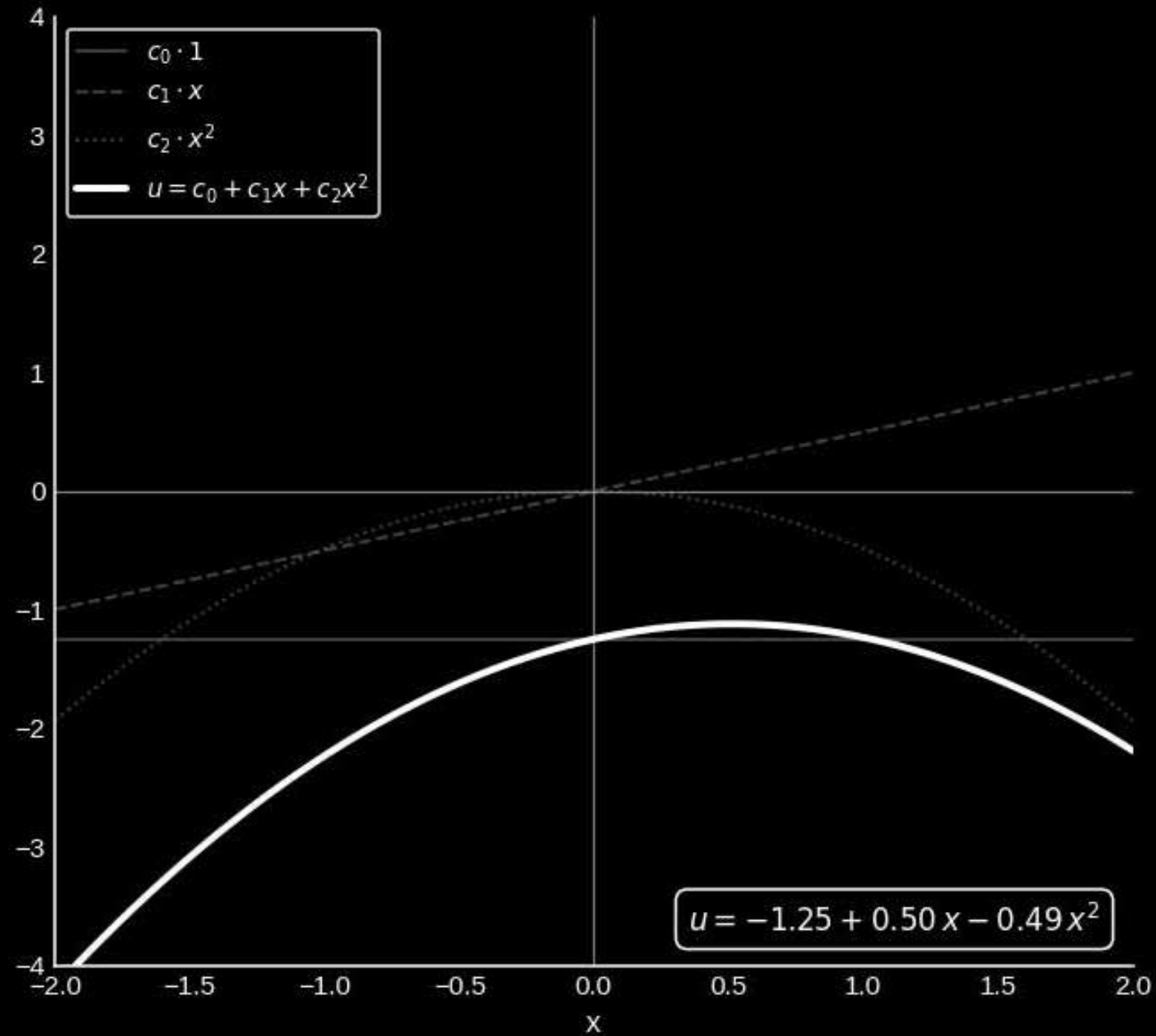
This establishes an **isomorphism** $\phi : V \rightarrow \mathbb{F}^n$ defined by $\phi(u) = [c_1, \dots, c_n]^T$, proving that every n -dimensional vector space over $\mathbb{F}$ is isomorphic to $\mathbb{F}^n$.



Isomorphism $\phi: \mathcal{P}_2 \rightarrow \mathbb{R}^3$ with basis $\mathcal{B} = \{1, x, x^2\}$

Polynomial $u \in \mathcal{P}_2$

Coordinate vector $\phi(u) \in \mathbb{R}^3$



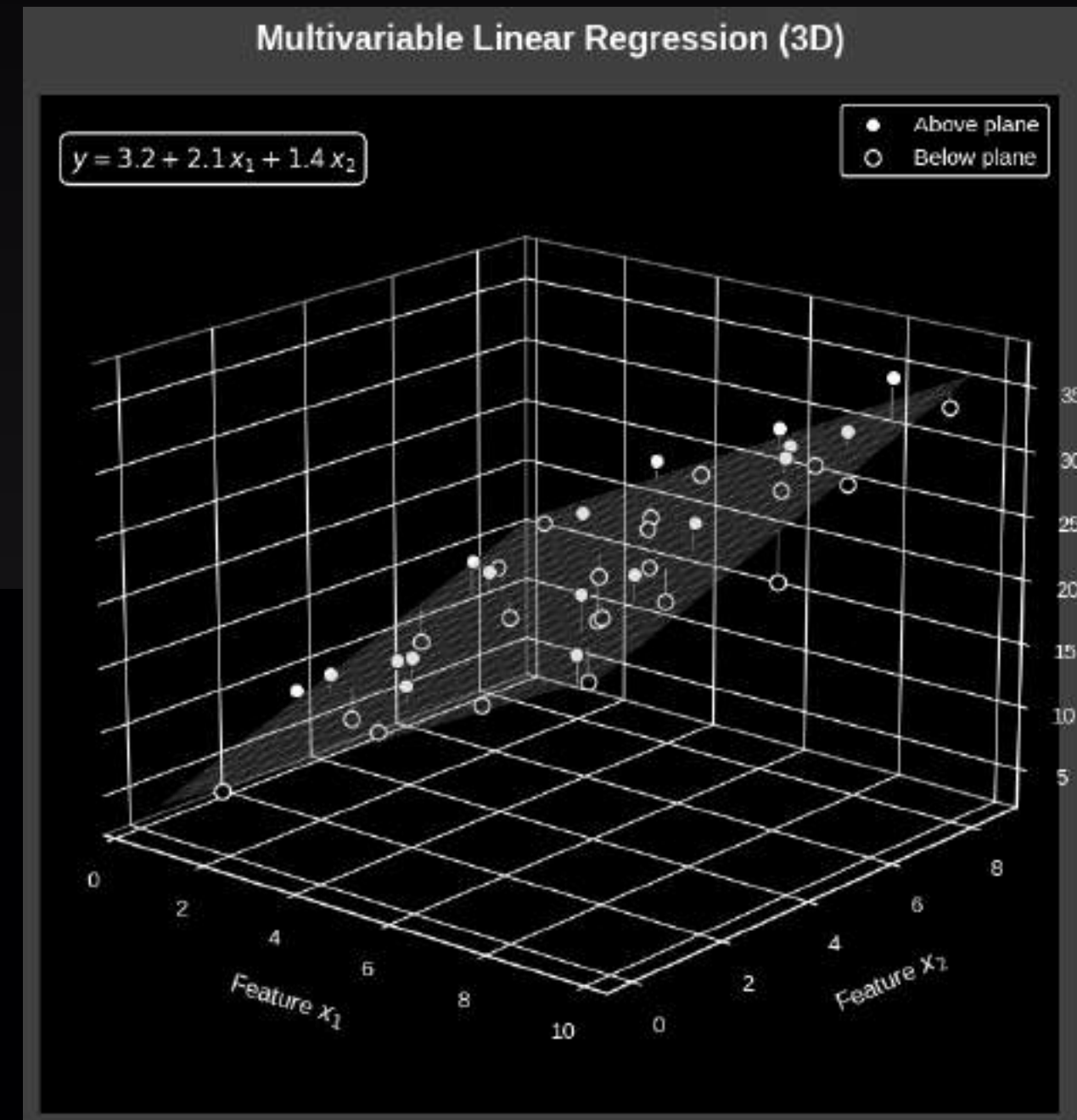
Multivariate linear regression

$$\hat{y} = m_1x_1 + m_2x_2 + \dots + m_nx_n + b$$

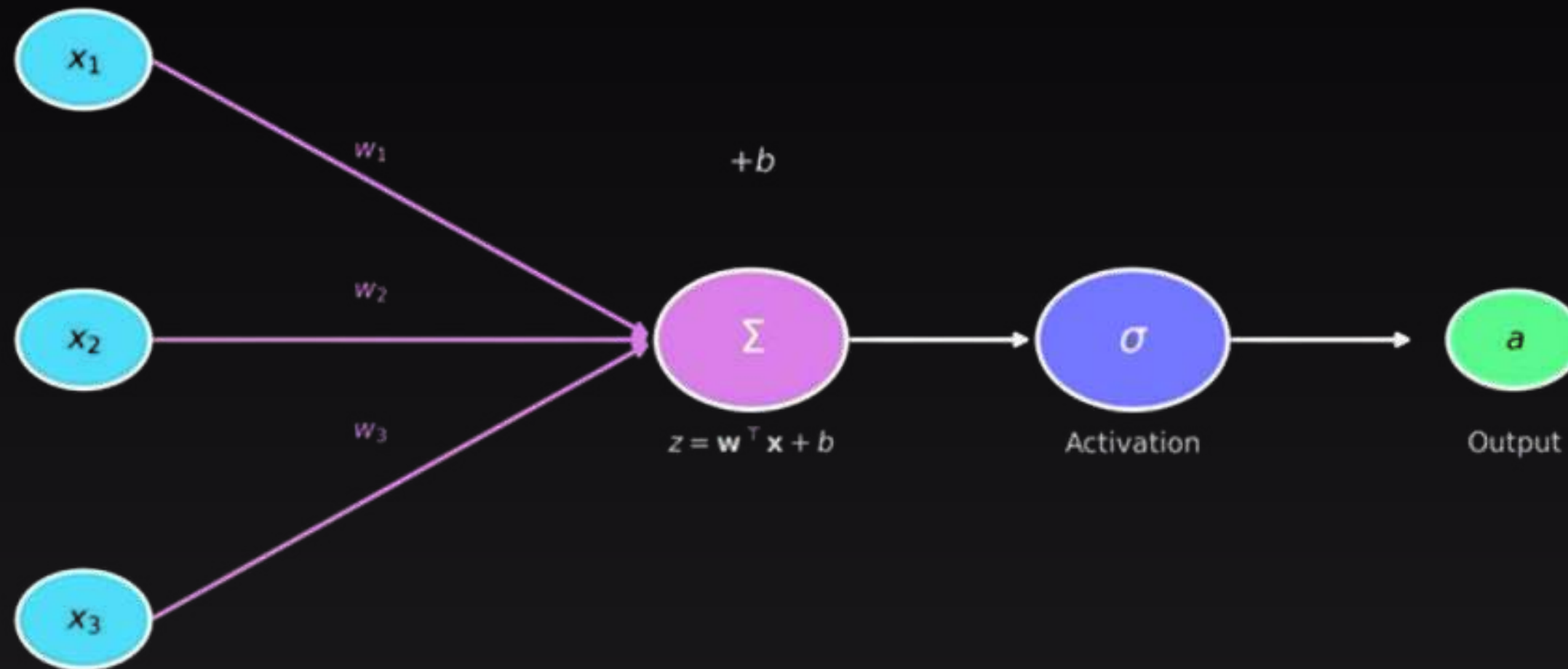
$$\hat{y} = W^T x + b$$

$$\hat{y} = X \cdot w$$

It is an extension of simple linear regression. While in simple linear regression we use a single feature to make predictions, in multivariate regression we use two or more independent variables to predict a single value.



Is It a single Neuron a Linear Regression?



References

Marsland, S. (2015). Machine learning: An algorithmic perspective (2.^a ed.). CRC Press.
https://mitmecsept.wordpress.com/wp-content/uploads/2018/05/chapman-hall_crc-machine-learning-pattern-recognition-stephen-marsland-machine-learning_-an-algorithmic-perspective-second-edition-2014-chapman-and-hall_crc.pdf